



Course: MCA (A.Y-II) 2020

Semester: 3

Prerequisite: Fundamental knowledge of Networking and Computing, Basic Knowledge of Web Services

Course Objective: The key objectives of this course are to provide an understanding of the basic concepts of parallel and distributed computing and their role in cloud computing, to study the concept of virtualization and relevant technologies available in the market, to understand the importance of cloud computing for higher throughput, to make aware about availability of various cloud platforms, to study different application of cloud and cloud management techniques.

Teaching and Examination Scheme										
Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	1	2	—	5	20	20	20	60	30	150

SEE - Semester End Examination, CIA - Continuous Internal Assessment (It consists of Assignments/Seminars/Presentations/MCQ Tests, etc.)

Course Content	W - Weightage (%) , T - Teaching hours		
Unit No	Revised Syllabus Content	W	T
1	Cloud Principles & Well-Architected Framework Fundamental Principles of Cloud Computing, Shared Responsibility Model in Cloud Understanding Cloud-Native Architecture, Well-Architected Framework (Security, Reliability, Performance, Cost, and Sustainability), Best Practices for Designing Scalable Cloud Applications	15	7
2	Disaster Recovery in Cloud Concepts of Backup & Disaster Recovery, High Availability & Fault Tolerance in Cloud Disaster Recovery Strategies (RPO & RTO Considerations), Cloud-Based Backup Solutions & Snapshots Case Studies on Disaster Recovery Implementation	15	7
3	Introduction to Infrastructure as Code (IaC) Overview of Infrastructure as Code (IaC), Tools for IaC: Terraform, AWS CloudFormation, Ansible Configuration Management & Automation, Writing & Managing Infrastructure as Code, Best Practices for Versioning & Scaling Infrastructure	20	10
4	Kubernetes & Container Orchestration Introduction to Kubernetes & its Architecture, Setting Up & Managing Kubernetes Clusters, Deploying Applications on Kubernetes, Kubernetes Networking & Load Balancing, Monitoring & Scaling Kubernetes Workloads	15	7
5	Introduction to CI/CD Understanding DevOps and CI/CD Concepts, CI/CD Pipelines: Overview & Implementation, Tools for CI/CD: Jenkins, GitHub/Git Co-Pilot Actions, GitLab CI, Automated Testing & Deployment, strategies, Best Practices for Secure & Scalable CI/CD Pipelines	15	7



6	Cloud Application Integration Understanding Cloud Application Integration Concepts, API Gateway & Microservices Communication Serverless Functions & Event-Driven Architecture, Cloud Messaging Services (Pub/Sub, Kafka, RabbitMQ), Real-World Use Cases of Cloud Integration	20	10
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Reference Books	
1.	Cloud Computing - A practical approach for learning and implementation By A.Srinivasan and J.Suresh Pearson Publications
2.	Cloud Computing: A practical approach By Anthony T. Vetle TMH
3.	Cloud Computing For Dummies By Judith Hurwitz, Robin Bloor, Marcia Kaufman, Fern Halper Wiley India Pvt Ltd
4.	Cloud Computing: SaaS, PaaS, IaaS, Virtualization, Business Models, Mobile, Security and More By Kris Jamsa Jones & Bartlett Learning
5.	Cloud Computing Bible By Barrie Sosinsky Wiley India Pvt Ltd

Course Outcome
After Learning the Course the students shall be able to: • define cloud computing and its applications. • describe role of virtualization in establishing cloud. • understand principles and service models of cloud computing. • familiarize with various cloud computing platforms. • conceptualize cloud storage management and data security. • identify feasible cloud platforms for solutions to be provided. • deploy services and resources on cloud platforms.

PRACTICALS	
Pract No	Revised PRACTICALS LIST
1	Creating a Google Cloud Free Tier Account. Deploying a Linux VM on GCP
2	Deploying a Docker Container on GCP Compute Engine
3	Deploying a Containerized Application on Cloud Run
4	Setting Up Kubernetes Cluster and Deploying & Scaling an App on Kubernetes
5	Implementing Cloud Storage Backup & Snapshots
6	Creating a Cloud Function for Serverless Execution
7	Automating Infrastructure Deployment with Terraform
8	Implementing a CI/CD Pipeline with Cloud Build
9	API Gateway & Cloud Run for Microservices Communication and Cloud Pub/Sub for Event-Driven Messaging
10	Monitoring & Logging Kubernetes Workloads
11	Implementing a Disaster Recovery Plan (RPO & RTO Demo)